HAna Note

1 Basic Knowledge & FPM

1.1 Definitions, Useful Tools & Nested Intervals

Def of $\lambda(I)$: $\lambda((a,b)) = \lambda([a,b)) = \lambda((a,b]) = \lambda([a,b]) = b - a$. · **Def of** $\chi_I(x)$: $\chi_I(x) := \mathbb{1}_{x\in I}$

Def of limit superior and inferior: $\limsup a_n := \lim(sup_{k\geq n} a_k)$ $\liminf a_n := \lim(inf_{k\geq n} a_k)$

Important Inequation: $|x+y| \leq |x| + |y|$; $|x-y| \geq |x| - |y|$; $||x| - |y|| \leq |x \pm y|$. $|sin(x)| \leq |x|$ ps:need prove,By MVT

Bernoulli's Inequality: If $\alpha > 0, \delta \geq -1$. Then $(1+\delta)^\alpha \geq 1 + \delta\alpha, \alpha \in [1, +\infty)$

Def of Nested Interval: Sequence $(I_n)_{n\in\mathbb{N}}$ is nested if $I_n \supseteq I_{n+1}$ for all $n \in \mathbb{N}$. · **MCT**: $UB \Rightarrow sup$; $LB \Rightarrow inf$.

Nested Interval Theorem: If $(I_n)_{n\in\mathbb{N}}$ is nested closed bounded interval sequence $\Rightarrow E := \cap_n I_n \neq \emptyset$. If $\lim \lambda(I_n) = 0 \Rightarrow E = a, a \in \mathbb{R}$

Power Set $\mathcal{P}$: $P(E)$ is the set of all subsets of E. ps: $\mathcal{P}(\mathbb{R})$ is the set of all subsets of $\mathbb{R}$.

Countable Set: A set E is countable if $\exists$ bijection $f : \mathbb{N} \to E$.

· Properties: **1**. Countable union of countable sets is countable. **2**. Product of finite countable sets is countable. **3**. Addition of finite countable sets is countable.

· **Exmaple**: $[0,1]$ is uncountable.

1.2 Infinite Sequence, Cauchy Sequence and Bolzano-Weierstrass Theorem

Useful Seq Limit: $\lim \frac{1}{n^p} = 0, p > 1$; $\lim a^n = 0, |a| < 1$; $\lim n^{\frac{1}{n}} = 1$; $\lim a^{\frac{1}{n}} = 1, a > 0$

Subsequence Theorem: In each case, the subsequence can be taken to be monotonic.

1. $\exists a \in \mathbb{R}\ s.t. \forall\varepsilon > 0$, there are infinitely many $n \in \mathbb{N}\ s.t. |x_n - a| < \varepsilon \Leftrightarrow$ $\exists$ subsequence of (x_n) $s.t.$ $\lim\ x_{n_k} = a$
2. $(x_n)_{n\in\mathbb{N}}$ is not bounded above (below) $\Leftrightarrow$ $\exists$ subsequence of x_n $s.t.$ $\lim x_{n_k} = \infty\ (-\infty)$
3. If $\lim x_n = L \Rightarrow \forall x_{n_k}, \lim x_{n_k} = L$

Cauchy Sequence: Sequence $(a_n)_{n\in\mathbb{N}}$ is Cauchy iff $\forall\varepsilon > 0, \exists N \in \mathbb{N}\ s.t. |a_n - a_m| < \varepsilon$ for all $m, n \geq N$. · **Convergent** $\Leftrightarrow$ **Cauchy**

Bolzano-Weierstrass Theorem: Every *bounded sequence* has a *convergent subsequence*. ps: In real number.

1.3 Infinite Series and their theorems

· **p-Test**: $\sum \frac{1}{n^p}$ convergent iff $p > 1$ **Root Test** **Ratio Test**$_{|n|^{1/n}, |n+1/n|}$ **Geometric Series**: $\sum p^n$ converges iff $|p| < 1$

Divergence Test: If $\lim a_n \neq 0 \Rightarrow \sum a_n$ diverges. (If $\sum a_n$ convergent $\Rightarrow \lim a_n = 0$.)

Comparison Test: If $0 < a_n < b_n, \sum b_n$ convergent $\Rightarrow \sum a_n$ also ; $\sum a_n$ divergent $\Rightarrow \sum b_n$ also.

· **Limit Comparison Test**: If $a_n, b_n \geq 0, \lim \frac{a_n}{b_n} = L > 0$ $\Rightarrow$ If $L \in (0,\infty)$: $\sum a_n$ convergent iff $\sum b_n$ also.
If $L = 0 : \sum b_n$ convergent $\Rightarrow \sum a_n$ also. $L = \infty : \sum b_n$ divergent $\Rightarrow \sum a_n$ also.

Integral Test: Let $f : [1,\infty) \to \mathbb{R}$ is 非负递减, $a_n = f(n)$. Then $\sum a_n$ converges iff $\int_1^\infty f(x)dx < \infty$.

Absolutely Convergence: $\sum a_n$ convergent absolutely iff $\sum |a_n|$ convergent. **If convergent abs $\Rightarrow$ convergent.**

Alternating Series Test: If a_n decreasing, $a_n \geq 0, \lim a_n = 0$. Then $\sum(-1)^{n-1} a_n$ convergent.

Cauchy's Condensation Test: If $a_n \geq 0, a_n$ decreasing, $\Rightarrow$ $[\sum a_n convergent \Leftrightarrow \sum 2^n a_{2^n}\ also\ ]$

Cauchy Criterion for Series: $\sum_{k=1}^\infty a_k$ converges iff $\forall\varepsilon > 0, \exists N \in \mathbb{N}\ s.t. \forall n \geq m \geq N\ |\sum_{k=m+1}^n a_k| = |a_{m+1} + \cdots + a_n| < \varepsilon$.

Riemann Rearrangement Theorem: If $\sum_{k=1}^\infty a_k$ is conditionally convergent. $\Rightarrow$ $\forall s \in \mathbb{R}, \exists$ bijection $\sigma : \mathbb{N} \to \mathbb{N}$ s.t. $\sum_{n=1}^\infty a_{\sigma(n)} = s$.

Rearrangement for Abs Series: If $\sum_{n=1}^\infty a_n$ is absolutely convergent. $\Rightarrow$ $\forall$ bijection $\sigma : \mathbb{N} \to \mathbb{N}$, $\sum_{n=1}^\infty a_{\sigma(n)} = \sum_{n=1}^\infty a_n$.

1.4 Continuous, differentiable functions and their theorems

Def of lim of func: **1**.$\forall\varepsilon > 0, \exists\delta > 0\ s.t. 0 < |x - x_0| < \delta, \Rightarrow |f(x) - L| < \varepsilon$. **2**.$\forall x_n \neq x_0\ s.t. \lim(x_n) = x_0, then\ \lim f(x_n) = L$.

Def of cont of func: **1**.$\forall\varepsilon > 0, \exists\delta > 0\ s.t. \forall x, |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$. **2**.$\forall(x_n)_{n\in\mathbb{N}}, if\ \lim x_n = x_0 \Rightarrow \lim f(x_n) = f(x_0)$

Def of diffi: **1**.$f'(a) = \lim\limits_{x\to a} \frac{f(x)-f(a)}{x-a}$ **2**.$f'(a) = \lim\limits_{h\to 0} \frac{f(a+h)-f(a)}{h}$.

The Extreme Value Theorem: If $f : [a,b] \to \mathbb{R}$ is continuous, then f has a max and min value.

Intermediate Value Theorem: $f : [a,b] \to \mathbb{R}$ is continuo,$f(a) \neq f(b), y \in [f(a), f(b)]$.$\Rightarrow \exists c \in (a,b)\ s.t. f(c) = y$

IVT (for deriv) / (Darboux Theorem): If f differentiable in $I \Rightarrow \forall m \in (f'(a), f'(b))$ we have: $\exists c$ s.t. $f'(c) = m$

Rolle's Theorem: If f continuous on $[a,b]$, differentiable on (a,b) and $f(a) = f(b)$ Then: $\exists c \in (a,b)$ s.t. $f'(c) = 0$

Mean Value Theorem: Let f, g continuous on $[a,b]$ and differentiable on (a,b): ps: a=b 时依然成立

· **1**.$\exists c \in (a,b)$ s.t. $\frac{f(b)-f(a)}{b-a} = f'(c)$ **2**. $\exists c \in (a,b)$ s.t. $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$

Theorem: If f is 1-1 and continuous $\Rightarrow$ f is strictly monotone. and f^{-1} is continuous and strictly monotone.

Inverse Function Theorem: Let f differentiable in I (open interval), and $\forall x, f'(x) \neq 0$:

· **1.** $f(I)$ is open interval. **2.** f is invertible. **3.** f^{-1} is differentiable and $(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$

Second Deriv Test: f twice continuously differentiable in $I, f'(x_0) = 0$: **1.** $f''(x_0) > 0\ (< 0)$ local min (max) point.

2 Uniformly & Pointwise Convergence

2.1 Pointwise Convergence

Def of Pointwise Convergence: **1**.$f_n : E \to \mathbb{R}$ converge pointwise iff $\forall x \in E, \forall\varepsilon > 0, \exists N_{\varepsilon,x}\ s.t. \forall n > N\ |f_n(x) - f(x)| < \varepsilon$ **2**.$\forall x \in E, \lim f_n(x)$ exists **3**.$f(x) = \lim f_n(x)$

· pointwise convergence do NOT preserve: continuity, differentiability, $(\lim f_n)' \neq \lim f_n'$, $\int \lim f_n \neq \lim \int f_n$

2.2 Uniformly Convergence

Def of Graph: $\Gamma(f) := \{(x, f(x))|x \in E\}$ ps:$f: E \to \mathbb{R}$

Def of Vertical Neighbourhood $\mathcal{N}(f,\varepsilon) := \{(x,y)|x \in E, |y - f(x)| < \varepsilon\}$ ps:$f: E \to \mathbb{R}$

Def of Uniformly Convergence: $f_n : E \to \mathbb{R}$ converge uniformly iff $\forall \varepsilon > 0, \exists N_\varepsilon\ s.t.\ \forall n > N, \forall x \in E, |f_n(x) - f(x)| < \varepsilon$ **Equvialent to**

1. $\lim \left[\sup_{x\in E} |f_n(x) - f(x)|\right] = 0$ ·**Uniformly** $\Rightarrow$ **Pointwise**
2. **Vertical Neighbourhood Test**: $\forall \varepsilon > 0, \exists N \in \mathbb{N}\ s.t.\ \forall n > N, \Gamma(f_n) \subset \mathcal{N}(f,\varepsilon)$

Uniformly Cauchy: $\forall \varepsilon > 0, \exists N \in \mathbb{N}\ s.t.\ \forall x \in E, m, n > N\ |f_n(x) - f_m(x)| < \varepsilon$ ps: $f_n : E \to \mathbb{R}$ · **Converge uniformly** $\Leftrightarrow$ **Uniformly Cauchy**

Sequence Test: If $f_n : E \to \mathbb{R}$ converge uniformly to f. $\Rightarrow$ $\forall (x_n)_{n\in\mathbb{N}} \in E, \lim |f_n(x_n) - f(x_n)| = 0$ 测试不是时可以用

Property of Uniformly Convergent: Let $f_n, f, g : I \to \mathbb{R}$ Then: NOT preserve: differentiable. ps: 2. **Termwise differentiation**

1. If [1] $\lim f_n(x) = f(x)$ uniformly, [2] $\forall f_n$ continuous at x_0 $\Rightarrow$ f continuous at x_0. $\Downarrow \forall x \in I, (\lim f_n)'(x) = \lim f_n'(x)$
2. If [1] I is <u>bounded open</u>; [2] $\forall f_n$ diff, $\lim f_n' = g$ unif; [3] <u>$\exists x_0 \in I\ s.t.\ f_n(x_0)$ converges.</u> $\Rightarrow$ <u>$\lim f_n = f$ unif</u>; f' diff and $f' = g$.
3. Cannot use nowIf [1] $\lim f_n(x) = f(x)$ uniformly on *closed interval* $[a,b]$, [2] f_n integrable on $[a,b]$ $\Rightarrow$ $\lim \int_a^b f_n(x)dx = \int_a^b \lim f_n(x)dx$

Def of Series of Functions: Let $f_n : E \to \mathbb{R}$; Def partial sum $s_n(x) := \sum_{k=1}^n f_k(x)$

1. $\sum f_n$ converge pointwise iff s_n converge pointwise. 2. $\sum f_n$ converge uniformly iff s_n converge uniformly. 3. $\sum f_n$ converge absolutely (pointwise) iff $\sum |f_n|$ converge (pointwise).

Weierstrass M-Test: $f_n : E \to \mathbb{R}, \exists (M_n)_{n\in\mathbb{N}} \geq 0$ s.t. [1] $\forall x \in E, \forall n \in \mathbb{N} : |f_n| < M_n$; [2] $\sum M_n$ converges. $\Rightarrow$ $\sum f_n$ converge absolutely & Uniformly.

3 Power Series

3.1 Introduction, Radius of Convergence, Continuity and Differentiability of Power Series

Power Series: A series from $\sum_{n=0}^\infty a_n(x-c)^n$ **Center**: c. **Coefficients**: a_n.

Radius of Convergence (R): For a power series $\sum a_n(x-c)^n$, there are three situations for it's radius of convergence:

1. $R = 0$ if $\sum a_n(x-c)^n$ converges only $x = c$ || $R = \infty$ if $\sum a_n(x-c)^n$ converges absolutely $\forall x \in \mathbb{R}$
2. $\exists R > 0$ s.t. $\sum a_n(x-c)^n$ converges absolutely if $|x-c| < R$; diverges if $|x-c| > R$. || $R = \sup\{r \geq 0 : (a_n r^n)_{n\in\mathbb{N}}$ <u>*bounded*</u>$\}$

Theorem (Continuity, Uniformly and Differentiability): If $f(x) = \sum a_n(x-c)^n$ with $R > 0$.

1. $f(x)$ continuous on $(c-R, c+R)$
2. $\forall 0 < r < R, f(x)$ converges absolutely and uniformly on $[c-r, c+r]$
3. $\sum_{n=0}^\infty a_n(x-c)^n$ and $\sum_{n=1}^\infty na_n(x-c)^{n-1}$ have same R.
4. **Termwise differentiation**: $f(x)$ is differentiable on $(c-R, c+R)$ || $f'(x) = \sum_{n=1}^\infty na_n(x-c)^{n-1}, x \in (c-R, c+R)$
   $\forall 0 < r < R, f(x), f'(x)$ converges absolutely and uniformly on $[c-r, c+r]$
5. $f(x)$ is infinitely convergence on $(c-R, c+R)$ || $a_k = \frac{f^{(k)}(c)}{k!}$

3.2 The Exponential Function & Real Analytic Function

The Exponential Function: $E(x) := \sum_{n=0}^\infty \frac{x^n}{n!}$ with $R = \infty$ ps: usually denoted by $exp(x)$ or e^x

Real Analytic Function: $f : I \to \mathbb{R}$ is real analytic iff $\exists (a_n)_{n\in\mathbb{N}}$ s.t. $f(x) = \sum_{n=0}^\infty a_n(x-c)^n$ for $\forall x \in I$ i.e. Series converges pointwise on I.

Taylor Series: Given [1] *Open Interval* I, [2] *infinitely differentiable* $f : I \to \mathbb{R}$. $\Rightarrow$ Taylor Series of f at $c \in I := \sum_{n=0}^\infty \frac{f^{(n)}(c)}{n!}(x-c)^n$

· Not all infinitely differentiable functions are real analytic: e.g. $f(x) = \{e^{-1/x^2}, x > 0; 0, x < 0$

Taylor's Theorem: If $f : (a,b) \to \mathbb{R}$ is differentiable function at $x_0 \in (a,b)$:

1. $P_{n,f,x_0}(x) := f(x_0) + \sum_{k=1}^n \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$ (Taylor's polynomial of degree n at x_0)
2. If $f^{(n+1)}$ exists on (a,b), then: $\exists \xi \in (x, x_0)$ s.t. $f(x) = P_{n,f,x_0}(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-x_0)^{n+1}$
3. Remainder $R_{n,f,x_0} := f(x) - P_{n,f,x_0}$. || $f : I \to \mathbb{R}$ is real analytic iff $\lim R_{n,f,x_0} = 0$ for $\forall x \in I$

4 Integration Theory & Lebesgue Integration

4.1 Introduction

4.1.1 Definition of Integral and Step Function

Types of Integration & Fundamental Theorem of Calculus:

1. **Indefinite integrals**: Also called *primitives* or *anti-derivatives*. ps: 不定积分. || Definition: 求导的逆运算 || e.g. $\int f'(x)dx = f(x) + c$
2. **Definite integrals**: ps: 定积分 || Definition: 函数在指定区域下的面积 || $\int_a^b f'(x)dx = f(b) - f(a)$
3. **Fundamental Theorem of Calculus**: $\int_a^b f'(x)dx = f(b) - f(a)$

Characteristic function χ_E: Given $E \subset \mathbb{R}, \chi_E(x) := \{1, x \in E;\ 0, x \notin E$. || ps: $\lambda(I) = \sup I - \inf I$ | If $I = I_1 \cup I_2, I_1 \cap I_2 = \emptyset \Rightarrow \lambda(I) = \lambda(I_1) + \lambda(I_2)$

Def of Simple Integral: **1.** $\int_\mathbb{R} \chi_I(x)dx := \lambda(I)$ **2.** $\int_\mathbb{R} c\chi_I(x)dx := c\lambda(I)$ **3.** $\int_\mathbb{R} \sum_j c_j\chi_{I_j}(x)dx = \sum_j c_j\lambda(I_j)$

Step Function: <u>non-negative</u> $f : \mathbb{R} \to [0,\infty)$ is a step function iff $f(x) = \sum_j c_j\chi_{I_j}(x)$ || Def: $\int_\mathbb{R} f(x)dx = \sum_j c_j\lambda(I_j)$

4.1.2 Open Set and Closed Set

Open Set: A set S is open iff $\forall x \in S, \exists r > 0,\ s.t.\ (x-r, x+r) \subseteq S$ ps: Collection of all open subsets of $\mathbb{R}$ is called *the Euclidean topology*

Structure of open sets: If S is a nonempty open set $\Leftrightarrow$ S = <u>countable</u> *union* of <u>disjoint</u> *open intervals.* ps: 即 open set = ∪ open intervals

∪ of open sets: Let $\mathcal{U}$ be collection of some open sets. (Can Infinite) $\cup_{S\in\mathcal{U}} S$ is open.

∩ of open sets: Let $\mathcal{U}$ be collection of some open sets. (Finite) $\cap_{S\in\mathcal{U}} S$ is open.

Connection with continuous function|open set: If $f : \underline{\mathbb{R}} \to \mathbb{R} \Rightarrow \{x \in \mathbb{R} : f(x) > \lambda\}$ is open.

Closed Set: A set $F \subseteq \mathbb{R}$ is *closed* iff $\mathbb{R} \setminus F$ is open.

Connection with continuous function|open set: If $f : \underline{\mathbb{R}} \to \mathbb{R} \Rightarrow \{x \in \mathbb{R} : f(x) \le \lambda\}$ is closed.

Theorem: If F is closed set, $(x_n)_{n\in\mathbb{N}} \in F$ convergent. $\Rightarrow$ $x = \lim x_n \in F$

4.2 Lebesgue Measures

Countably Additivity: $(E_n)_{n\in\mathbb{N}}$ disjoint sets, m be measure. $\Rightarrow$ Countably Additive if: $m(\cup_n E_n) = \sum m(E_n)$

Lebesgue Outer Measure: $m^*(E) := \inf\left\{\sum_{n=1}^{\infty} \lambda(I_n) \mid E \subseteq \cup_{n=1}^{\infty} I_n, I_n\ is\ a\ open\ interval\right\}$ ps: 本质是寻找到最小的区间可以覆盖 E, 用前面 λ 的定义来定义" 长度"

· **Lebesgue Outer Measure (Version 2)**: $m^*(E) = \inf\{\lambda(U), E \subseteq U, U\ is\ open\ set\}$ ps:$\lambda(U) = \sum \lambda(I_n)$

· **Special Case**: If $J \subseteq \mathbb{R}$ is interval. $m^*(J) = \lambda(J)$

· **Remark**: If it's impossible to find $(I_n)_{n\in\mathbb{N}}$ s.t. $\sum \lambda(I_n) < \infty$ $\Rightarrow$ $m^*(E) = \infty$ || Can interpret m^* as $m^* : \mathcal{P}(\mathbb{R}) \to [0, \infty)$

Property of Lebesgue Outer Measure: Let Set $E, F\ s.t.\ E \subseteq F \subset \mathbb{R}$. Emptyset: $\emptyset$

· **1. Null Empty Set**: $m^*(\emptyset) = 0$. **2. Monotonicity**: $m^*(E) \le m^*(F)$. **3. Countable Subadditivity**: $m^*(\cup_{n=1}^{\infty} E_n) \le \sum_{n=1}^{\infty} m^*(E_n)$.

Measure Zero: A set E is said to have *measure zero* if $m^*(E) = 0$. e.g. Cantor Set. ⇓

· Every countable subset of $\mathbb{R}$ has measure zero. || $E \subseteq \mathbb{R}$ countable, $\Rightarrow$ $E = \cup_{j=1}^{\infty}\{x_j\}$ (Union of countable singletons) || $\exists m^*(E) = 0, E\ uncountable$

4.3 Countable Additive for Lebesgue Measure

Def of $dist(E, F)$: $dist(E, F) := \inf\{|x - y| : x \in E, y \in F\}$

· **Lemma**: If $^1E, F$ are closed sets, $^2E \cap F = \emptyset$, ^3E bounded. $\Rightarrow$ $dist(E, F) > 0$

Theorem: If $dist(E, F) > 0$ $\Rightarrow$ $m^*(E \cup F) = m^*(E) + m^*(F)$.

5 Appendix

Trigonometric func: s+s=sc,s-s=cs,c+c=cc,c-c=-ss || e.g. $sinA + sinB = 2sin(\frac{A+B}{2})cos(\frac{A+B}{2})$ $sinAsinB = -\frac{1}{2}[cos(A+B) - cos(A-B)]$

hyperbolic func: $sinh(x) = \frac{e^x - e^{-x}}{2}$ $cosh(x) = \frac{e^x + e^{-x}}{2}$ $cosh^2(x) - sinh^2(x) = 1$ $[sinh(x)]' = cosh(x)$ $[cosh(x)]' = sinh(x)$

Common Int: $\int tan(x)dx = -ln|cos(x)|$; $\int cot(x)dx = ln|sin(x)|$; $\int sec(x)dx = ln|sec(x) + tan(x)|$; $\int csc(x)dx = ln|csc(x) - cot(x)| = ln|tan(\frac{x}{2})|$

Common Der: $[tan(x)]' = sec^2(x)$; $[cot(x)]' = -csc^2(x)$; $[sec(x)]' = sec(x)tan(x)$; $[csc(x)]' = -csc(x)cot(x)$; $[arctan(x)]' = \frac{1}{1+x^2}$.

· $1 + tan^2(x) = sec^2(x)$; $1 - csc^2(x) = cot^2(x)$; $csc^2(x) - cot^2(x) = 1$. $[arcsin(x)]' = \frac{1}{\sqrt{1-x^2}}$; $[arccos(x)]' = -\frac{1}{\sqrt{1-x^2}}$

$\mathbf{q^n - 1}$: $q^n - 1 = (q-1)(1 + q + q^2 + \cdots + q^{n-1})$ || $\mathbf{1 \pm x^3}$: $1 \pm x^3 = (1 \pm x)(1 \mp x + x^2)$

Summation By Parts: $\sum_{k=1}^{n} a_k(b_{k+1} - b_k) = (a_{n+1}b_{n+1} - a_1b_1) - \sum_{k=1}^{n}(a_{k+1} - a_k)b_{k+1}$, $\forall (a_k)_{k=1}^{n+1}, (b_k)_{k=1}^{n+1}$

$\Rightarrow \sum_{k=1}^{n} a_k b_k = a_{n+1}\sigma_{n+1} - \sum_{k=1}^{n}(a_{k+1} - a_k)\sigma_{k+1}$, $\forall (a_k)_{k=1}^{n+1}, (b_k)_{k=1}^{n+1}$ $\sigma_k := \sum_{j=1}^{k-1} b_j$

Common Power Series: $\sum_{n=0}^{\infty} x^n = (1-x)^{-1}$ $\sum_{n=1}^{\infty} nx^{n-1} = (1-x)^{-2}$ $\sum_{n=2}^{\infty} n(n-1)x^{n-2} = 2(1-x)^{-3}$ $R = 1$ for all.

$\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$ $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{x^n}{n} = ln(x+1)$ $\sum_{n=1}^{\infty} \frac{x^n}{n} = -ln(1-x)$ $R = \infty$ for all.

Sum of $\mathbf{n, n^2, n^3}$: $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$ $\sum_{k=1}^{n} k^3 = [\frac{n(n+1)}{2}]^2$

Useful Conterexample: $(1/2)^{\frac{1}{n}} \subseteq [1/2, 1)$

Countably Approximation Tools: $\sum_{n=1}^{\infty} 2^{-n}\varepsilon = \varepsilon$